

Summary - Employee Attrition Dataset

1. Project Overview

- The project's aim was to develop an ML model, which could predict whether an employee will stay or leave the company.
- The IBM HR Attrition Analytics dataset was used. It had 34 features/factors, 1 target (thing to be predicted, i.e., attrition) and details of 1470 employees(1470 rows).
- 3 ML models, which are best at such tasks, were trained on the given data and the best one was picked carefully and chosen for further analysis.
- Important factors, trends and suggestions were recorded.

2. Major Findings

- Overtime, Monthly Income, Stock Option Level and Job Level are the strongest factors which affect attrition rates.
- Sales Department has highest exit rate.
- Within Sales Department, Sales Representatives are most likely to leave.
- Newly joined candidates(<2 years) leave more than the ones who have worked >3-5 years in the company. This in most cases correlates with job level. Employees at level 1 leave more than employees at higher level.
- Monthly Income matters a lot, but is not the only factor affecting attrition.

3. Overall Model Performance (non-technical)

- 3 predictive models were trained on the given data.
- The model chosen as the best model has an overall prediction accuracy of 82%, meaning it correctly predicts whether an employee will leave or stay about 82% of the time.

4. Recommendations for HR

- Policy Changes:
 - Distribute workload evenly so that employees do not land in a situation to work overtime regularly. It is affecting exit rate the most.
 - Review and consider deserving employees with lower salary for an increment since salary is also reported as an important aspect.
 - Review and revise company's stock option level policy, deserving employees can be awarded with a higher level since it financially benefits them.
 - Focus on increasing engagement with newly joined candidates and the ones at job level "1". Introducing incentives, good mentorship, assigning interesting projects etc. can be done to achieve that, since newer employees leave more.
- Retention Conversations:

- With Sales Department can be held, to understand their issues, verify the working conditions etc.
- Specifically with Sales Representatives, they seem to leave more than any job role in the company.
- With newly joined candidates to understand what's making them leave so soon and create a plan of action accordingly.

5. Limitation

- As stated earlier, model correctly predicts only 82% of the time.
- It might say employee is not going to leave when they actually are or sometimes say employee will leave when in fact they will not.
- Model should be used as a decision-support tool alongside human judgement to improve prediction results and overall progress.